Ashwin Balaji Arun Kumar

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Location: Dortmund, Germany

Education

Fachhochschule Dortmund – University of Applied Sciences and Arts Dortmund, DE
Master of Engineering (M.Eng.) in Embedded Systems Engineering 10/2025 – Current
(Results pending for first semester)

Anna University Coimbatore, India
Bachelor of Engineering (B.E.) in Electrical and Electronics Engineering 08/2018 – 06/2022
GPA: 1.5 / 4.0 (German equivalent)

Skills

Programming Languages: C, C++, Python, Assembly (Fundamentals)
Hardware: TI C2000 (TMS320F), Renesas RL78, STM32 (F4 Series), ESP32, RP2040
Communication Protocols: I2C, UART, SPI, Modbus, RS485, MQTT, CAN(Basics)
Tools & Build Systems: Matlab/simulink, Git, SVN, Latex, Linker Scripts, Make, CMake, Makefiles
Quality & Standards: Static Analysis (Coverity), ASPICE (Basics), FMEA, IEC standards

Work Experience

Schneider Electric Mumbai, India
Deputy Manager – Firmware 08/2023 – 09/2025

- Led **3-20 HP** VFD firmware development on TI and RL78 MCUs, following the **V-Model** approach.
- **Architected** a modular application layer, decoupling control logic from hardware using **HAL** for portability.
- Developed a custom **P&O MPPT** algorithm, achieving **98%** tracking efficiency and stable water discharge.
- **Supported** control loop modeling in **Simulink** to validate PID stability and sensor fusion.
- Optimized short-circuit protection to **200ns** using Trip Zones, thereby reducing field failures.
- Integrated **static analysis** into a CI/CD pipeline, ensuring **10+** high-quality firmware releases.
- Implemented flash-based EEPROM emulation to replace EEPROM, significantly reducing unit **BOM cost**.

Graduate Engineer Trainee – Electromechanical 07/2022 – 08/2023

- Supported **6A – 100A RCBO** range lifecycle by managing Design changes and customer complaints.
- Improved Isolators thermal performance (IEC 60947-3) by achieving a **7°C reduction in temperature rise**.
- **Improved product reliability** by identifying potential failures and implementing root cause corrections.

Projects

i-Practace – Automatic Tennis ball throwing machine

- Secured funding of **EUR 9,900** (Rs.9,48 lakhs) grant from Dept. of Science & Technology, Govt of India
- Managed the overall POC Development, including hardware selection, AC/DC motor control, sensor integration, firmware and IoT development.

Patents and Publications

- **Automated Tennis game coaching device** - Indian Design Patent Number - 367774-001 (Journal No-20/2023)
- Comparative Analysis on Industrial IoT Communication Protocols and its Future Directives - IEEE XPLORE

Languages

German: A2 (Actively pursuing B1) **English:** C1